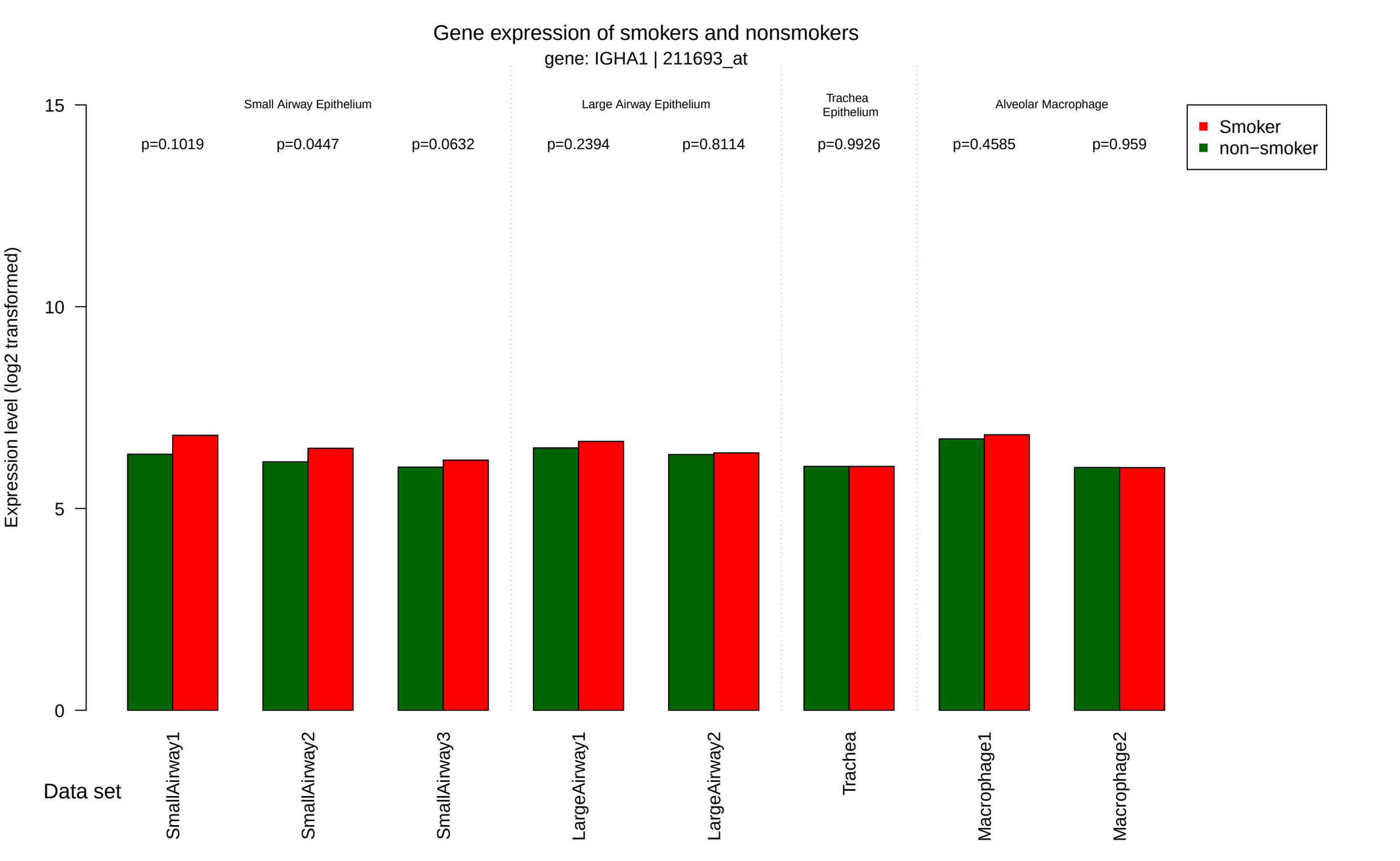
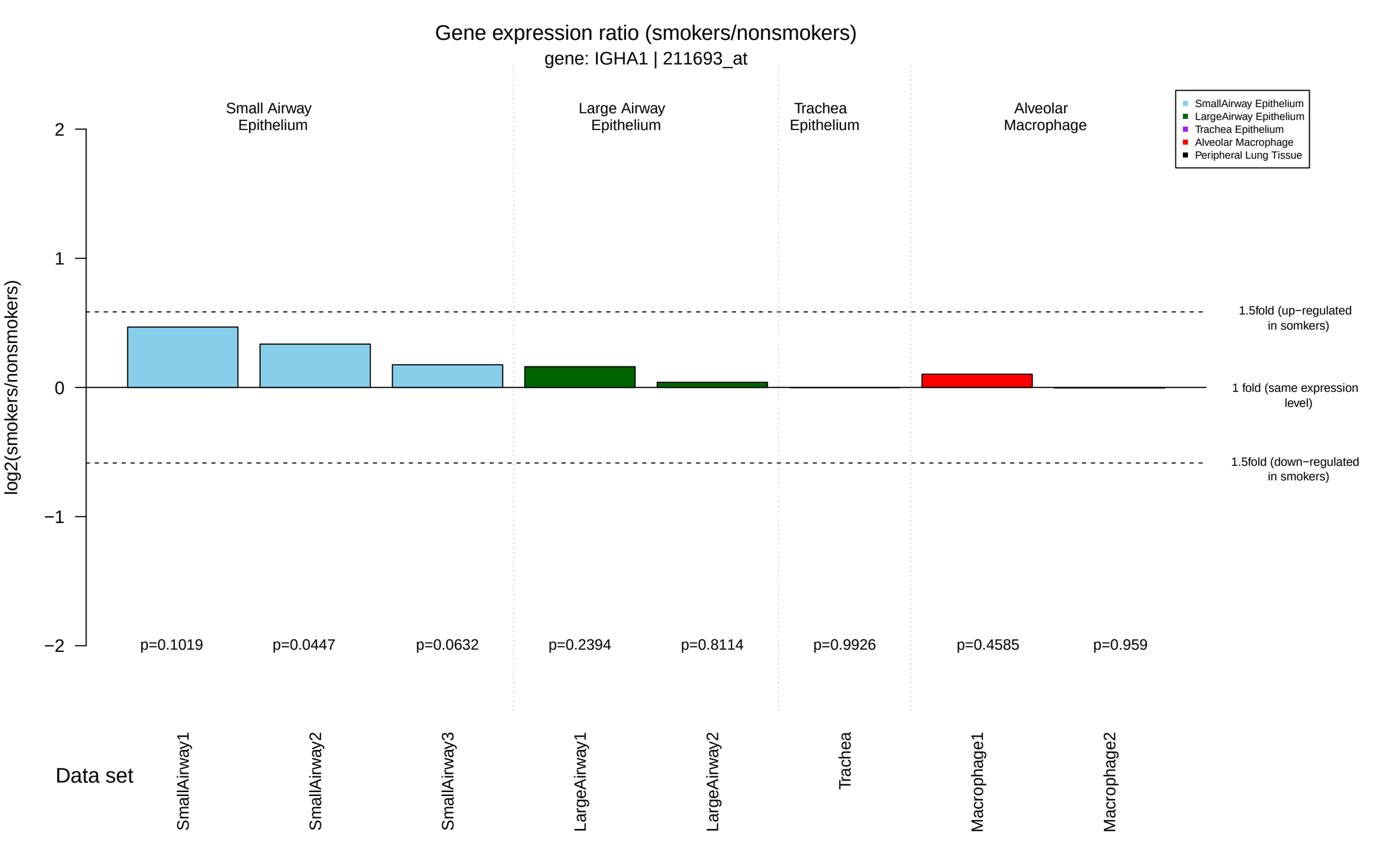
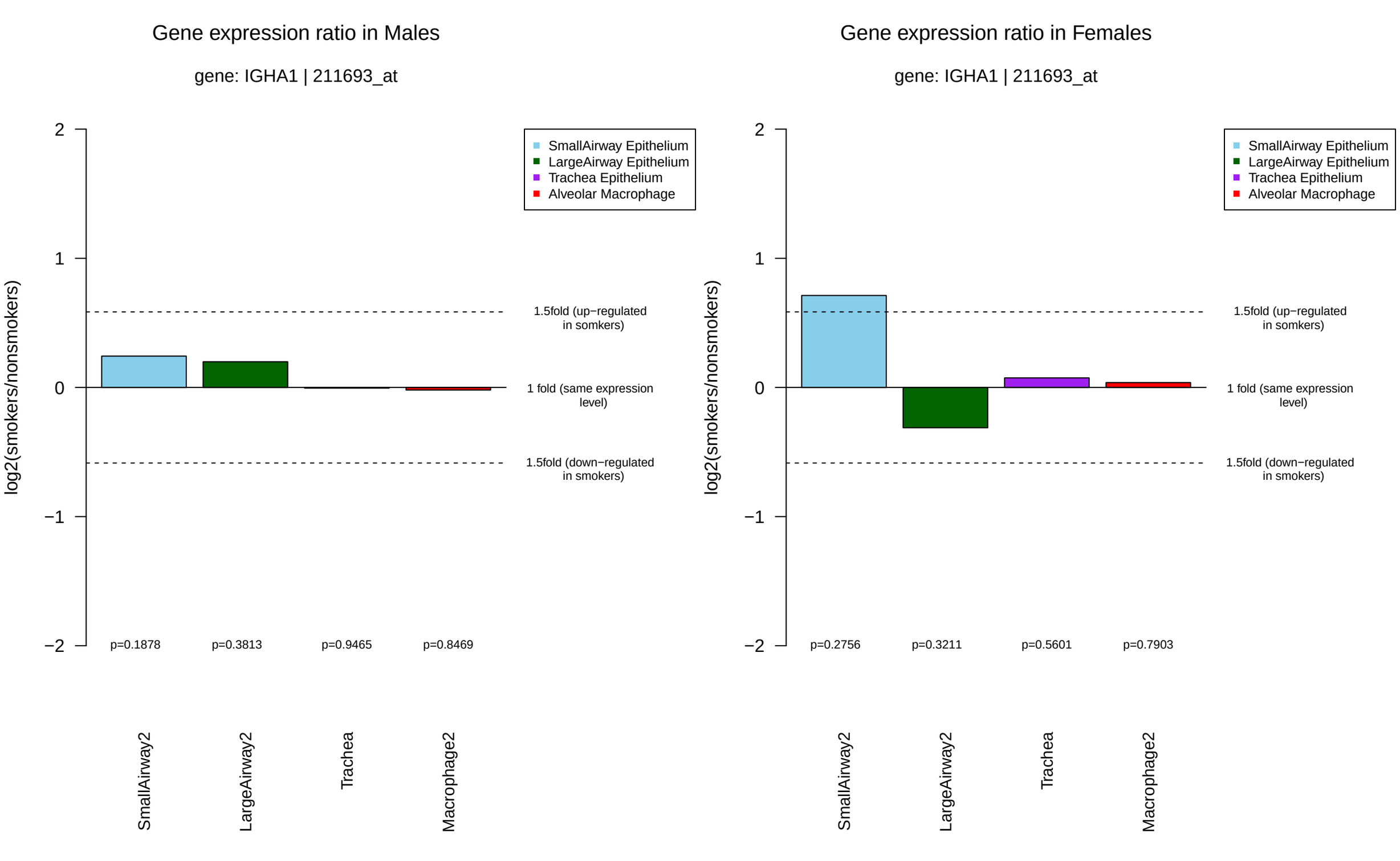
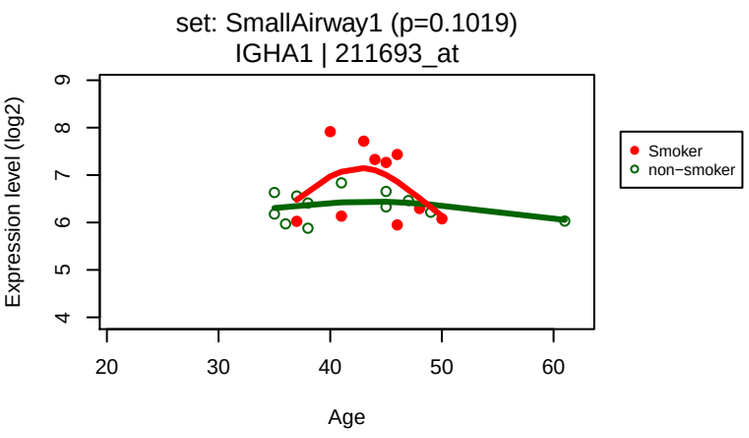
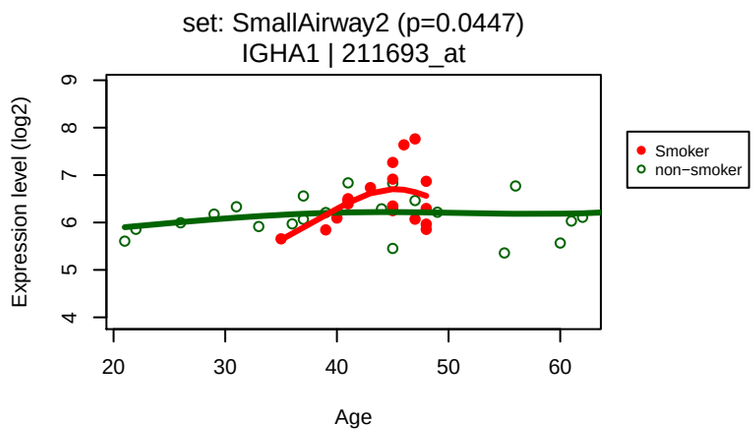
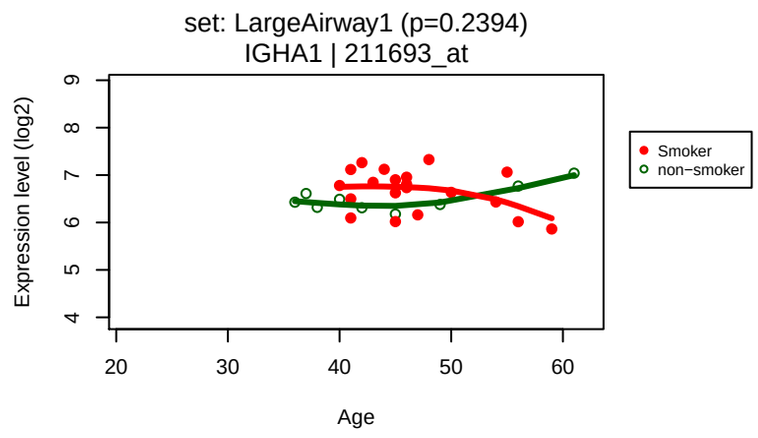

Gene expression of smokers and nonsmokers
gene: IGHA1 | 211693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1019
p=0.0447
p=0.0632
p=0.2394
p=0.8114
p=0.9926
p=0.4585
p=0.959
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

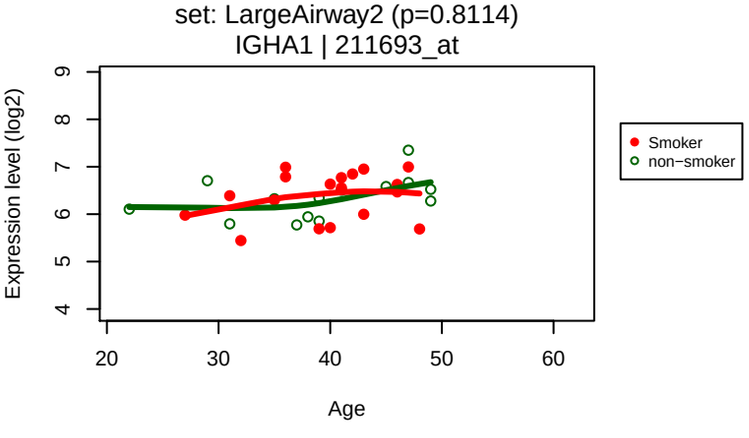

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 | 211693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1019
p=0.0447
p=0.0632
p=0.2394
p=0.8114
p=0.9926
p=0.4585
p=0.959
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

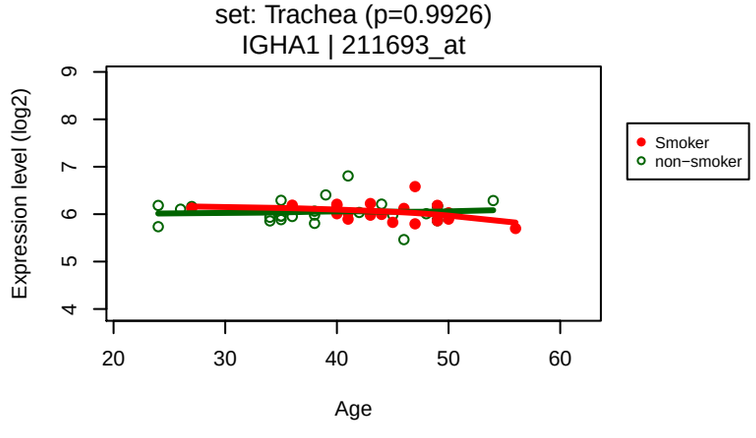

Gene expression ratio in Males
gene: IGHA1 | 211693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1878
p=0.3813
p=0.9465
p=0.8469
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 | 211693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2756
p=0.3211
p=0.5601
p=0.7903
SmallAirway2
LargeAirway2
Trachea
Macrophage2

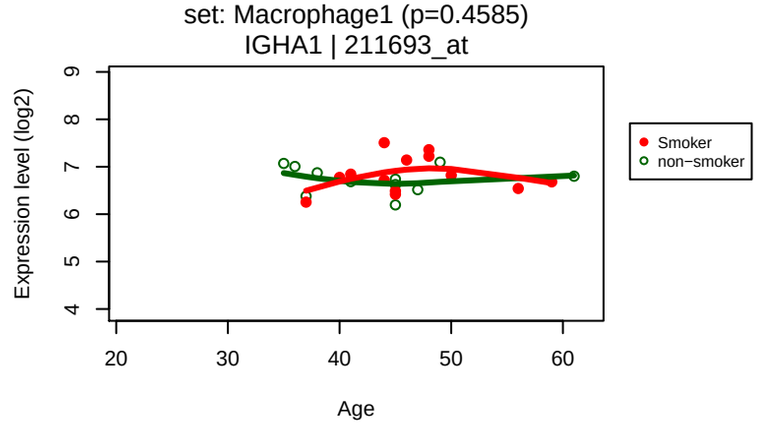

set: SmallAirway1 (p=0.1019)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker

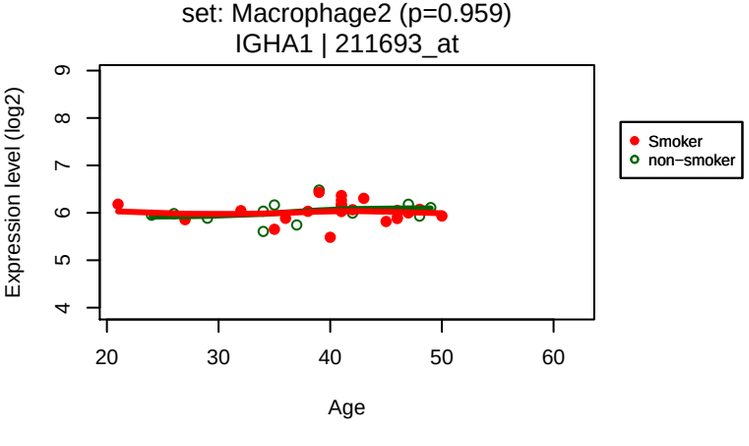

set: SmallAirway2 (p=0.0447)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker

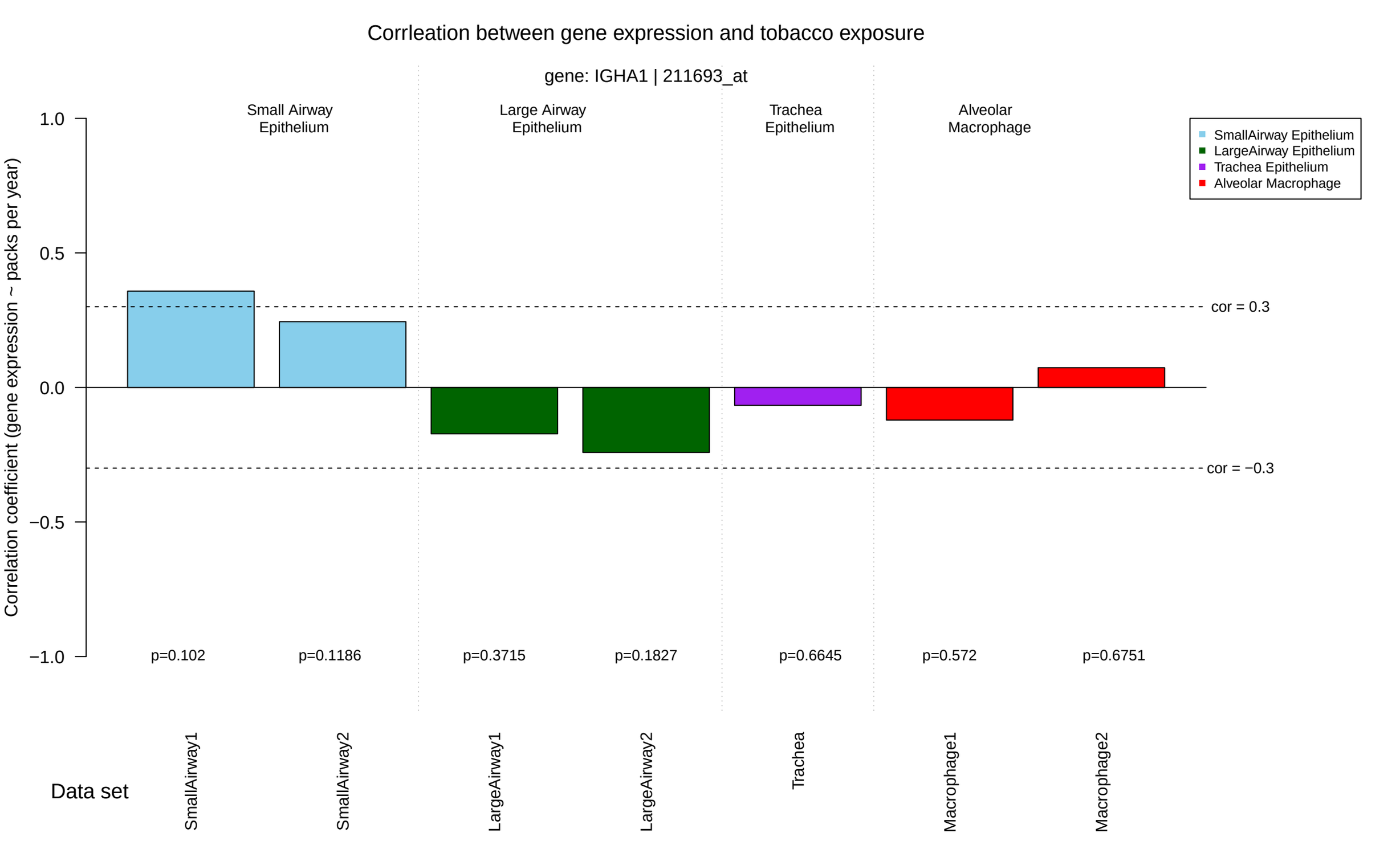

set: LargeAirway1 (p=0.2394)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8114)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9926)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4585)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.959)
IGHA1 | 211693_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 | 211693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.102
p=0.1186
p=0.3715
p=0.1827
p=0.6645
p=0.572
p=0.6751
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2